

Initial Post

◀ Initial Post

Initial Post ▶

Display replies in nested form

Settings ▾



Initial Post

by [Ali Alhammadi](#) - Wednesday, 6 August 2025, 6:44 PM

Agent-based systems (ABS) have become more popular in organizing intelligent, decentralized systems that coordinate complex and dynamic environments. Conventional systems are based on central control and fixed rules and usually do not work well in speed or scale (Axtell & Farmer, 2025). Conversely, ABS are independent agents who can perceive their surroundings, decide, and communicate with other agents to reach self- or group goals (Tour et al., 2021). The increase in ABS is deeply related to the progress in artificial intelligence. Such systems use methods of AI like machine learning, logic reasoning, and modeling behaviors. This merging enables the agents to behave independently, experience, and adapt to developments with no explicit human direction (Zolfagharipoor & Ahmadi, 2021). With the increasing size of technology and data, there has also been an increasing need to find systems that will deal with distributed processes, which makes ABS logical.

Agent-based systems bring several advantages to organizations. To begin with, they increase flexibility and scalability. Adding new agents can be done without reconfiguring the whole system, making expanding or changing operations easier. Second, they provide fault tolerance and strength, which is to say that should one of the agents malfunction, the rest can still operate. Third, AB facilitates decentralized decision-making, which minimizes bottlenecks and quick responses (Cincotti et al., 2022). Such systems can benefit supply chain management, robotics, financial markets, and healthcare, where autonomous elements must act in real time. Simulations can be performed using ABS to describe social, economic, or environmental systems, in order to understand emerging behavior and system-level outcomes. Examples of agent models include reactive agent models (responsive to instant inputs), deliberative agent models (plans using internally based goals), and hybrid models. Multi-agent systems take this further by allowing cooperation or competition among agents and resemble how human beings are organized (Rand & Stummer, 2021). In summary, ABS offers a dynamic, innovative business structure in a dynamic environment.

References

Axtell, R. L., & Farmer, J. D. (2025). Agent-based modeling in economics and finance: Past, present, and future. *Journal of Economic Literature*, 63(1), 197–287. Available at: <https://doi.org/10.1257/jel.20221319> (Accessed 6 August).

Cincotti, S., Raberto, M., & Teglio, A. (2022). Why do we need agent-based macroeconomics? *Review of Evolutionary Political Economy*, 3, 5–29. Available at: <https://doi.org/10.1007/s43253-022-00071-w> (Accessed 6 August).

Rand, W., & Stummer, C. (2021). Agent-based modeling of new product market diffusion: An overview of strengths and criticisms. *Annals of Operations Research*, 305, 425–447. Available at: <https://doi.org/10.1007/s10479-021-03944-1> (Accessed 6 August).

Tour, A., Polyvyanyy, A., & Kalenkova, A. (2021). Agent system mining: Vision, benefits, and challenges. *IEEE Access*, 9, 99480–99494. Available at: <https://doi.org/10.1109/ACCESS.2021.3095464> (Accessed 6 August).

Zolfagharipoor, M. A., & Ahmadi, A. (2021). Agent-based modeling of participants' behaviors in an inter-sectoral groundwater market. *Journal of Environmental Management*, 299, 113560. Available at: <https://doi.org/10.1016/j.jenvman.2021.113560> (Accessed 6 August).



Maximum rating: -

Permalink Reply

Chat to us!